macroscopic object. On the other hand, branching when decoherence occurs is already part of the Many-Worlds approach; all that this kind of spontaneous-collapse theory would say is “It’s just like Everett, except that when new worlds are created, we erase them by hand.” Who knows? That might be how nature actually works, but it’s not a route that most working physicists are encouraged to pursue.

Since the very beginning of quantum mechanics, an obvious possibility to contemplate has been the idea that the wave function isn’t the whole story, but that there are also other physical variables in addition to it. After all, physicists were very used to thinking in terms of probability distributions from their experience with statistical mechanics, as it had been developed in the nineteenth century. We don’t specify the exact position and velocity of every atom in a box of gas, only their overall statistical properties. But in the classical view we take for granted that there is some exact position and velocity for each particle, even if we don’t know it. Maybe quantum mechanics is like that—there are definite quantities associated with prospective observational outcomes, but we don’t know what they are, and the wave function somehow captures part of the statistical reality without telling the whole story.

We know the wave function can’t be exactly like a classical probability distribution. A true probability distribution assigns probabilities directly to outcomes, and the probability of any given event has to be a real number between zero and one (inclusive). A wave function, meanwhile, assigns an amplitude to every possible outcome, and amplitudes are complex numbers. They have both a real and an imaginary part, either one of which could be either positive or negative. When we square such amplitudes we obtain a probability distribution, but if we want to explain what is experimentally observed, we can’t work directly with that distribution rather than keeping the wave function around. The fact that amplitudes can be negative allows for the interference that we see in the double-slit experiment, for example.

There’s a simple way of addressing this problem: think of the wave function as a real, physically existing thing (not just a convenient summary of our incomplete knowledge), but *also* imagine that there are additional